Theory-Based User Modeling for Intelligent Tutoring Systems via Grounded Transformers

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Abstract. ~~The abstract should briefly summarize the contents of the paper in 150–250 words.~~The student model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these traditional approaches are subject to significant limitations. First, they lack pedagogical interpretability, as they are not anchored to theoretical frameworks derived from cognitive or educational principles. Furthermore, they frequently lack longitudinal context, as instructional decisions are typically based on isolated knowledge states without accounting for the historical trajectory that led there. To overcome these limitations, we propose a new approach based on Grounded Transformers that supports instructional decisions grounded in learning theories and individual learning histories, thus providing a theoretically sound foundation for TEL.

Keywords: deep knowledge tracing · theory-based interpretability · learning theories · intelligent tutoring systems · user modeling · grounded transformers.

1 Introduction

1.1 Problem Statement

The student model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these traditional approaches have one significant limitations: they lack pedagogical interpretability, as they are not anchored to theoretical frameworks derived from cognitive or educational principles.

To overcome this limitation, we propose a new approach based on Grounded Transformers that supports instructional decisions grounded in learning theories, thus providing a theoretically sound foundation for TEL.

1.2 Template Guidelines

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	**Lecture Notes**	14 point, bold
1st-level heading	**1 Introduction**	12 point, bold
2nd-level heading	**2.1 Printing Area**	10 point, bold
3rd-level heading	**Run-in Heading in Bold.** Text follows	10 point, bold
4th-level heading	*Lowest Level Heading.* Text follows	10 point, italic

y Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional)

acknowledgments[1], for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

References

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